Aaron Leevord Fernandes

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EDUCATION

Indiana University Bloomington Aug 2023 - May 2025
Master of Science in Intelligent Systems Engineering - Computer Engineering ***GPA: 3.9/4.0***

Goa College of Engineering Aug 2017 - Jul 2021
Bachelor of Engineering - Electronics and Telecommunication

EXPERIENCE

Machine Learning Engineer, Image Processing and Optics (Leung Research Group) Apr 2024 – Present
Biomedical Video Segmentation Project

- Designed deep learning CNN-GRU and Transformers based architectures in Pytorch for Optical Coherence Tomography (OCT) unsupervised video segmentation and spatio-temporal analysis to enable virtual histology (anomaly detection and tissue classification)
- Combined signal processing, Fourier-domain analysis, and machine learning, providing 50% better details than previous works.
- Solved tissue drift and warping during imaging by classical affine registration methods (Dipy) and voxel morph inspired U-Net based architectures for regularized diffeomorphic registration decreasing MSE by 10%.

Metasurfaces: High-Q Biomedical sensing Project

- Built High Performance Computing (Slurm, MPI) accelerated Lumerical FDTD simualtion pipelines for Meta-Surface design and synthetic data generation improving generation speeds by 80%.
- Engineered U-Net based FDTD solver surrogate models for forward and inverse metasurface design in tensorflow, reducing simulation runtime by 70% while maintaining 95% accuracy, achieving up to 15% gains over prior benchmarks.

Reinforcement Learning Engineer (Indiana University) Jan 2025 - May 2025
Multi Target Path Planning and Routing for Robotic Environments Project

- Identified limitations of value-based methods (e.g., DQN: 20% success) in stochastic, multi-target routing environments.
- Implemented and benchmarked policy-gradient algorithms (REINFORCE, PPO, TRPO, SAC, A2C) in PyTorch across custom tabular path-planning tasks with moving targets and obstacles; PPO achieved up to 90% success.
- Addressed sparse reward challenges by integrating Goal-Conditioned Discrete Actor-Critic with Stein Variational Goal Generation, GAN-based goal proposals, and Hindsight Experience Replay (HER), enabling successful completion of complex multi-goal tasks previously unsolvable under standard RL setups.

Software Engineer (Tata Consultancy Services - HDFC Bank) Jan 2022 - Jan 2023

- Developed robust financial transaction systems using Java, Spring Framework and Hibernate, improving processing efficiency
- Engineered SOAP/HTTP APIs and Microservices, enhancing system interoperability and reducing latency by 10-15%
- Streamlined transactional workflows, ensuring data consistency and seamless integration, reducing reconciliation errors by 7-15%

PROJECTS

Knowledge Graph Visualization App for Grounded Theory Analysis Oct 2024 - Dec 2024

- Developed an Electron-based cross-platform desktop application in Node.js and React for the NICC, Brussels.
- IInteractive relationship visualization and theory extraction using Graph based (Neo4j), LLM-powered NLP pipelines (LangChain and Llamaindex) to process qualitative interview data using the grounded theory framework.
- Targeted specifically the "coding", part of the framework, which generated about 90% codes as a human researcher. Reduced analysis timelines from months to days.

Efficient Model-Based Deep Reinforcement Learning With Predictive Control Oct 2024 - Dec 2024

- For better sample efficiency in RL methods, proposed and Implemented an MPC based RL framework with system Identification (Learned Model) and Value Iteration, converging in 200 episodes (best case) versus 1000+ for model-free methods.
- Learned dynamics model allowed the use of Cross Entropy Method for trajectory optimization and planning using the learning Model Network (Delta prediction network), increasing sample efficiency. (80% faster convergence)

Vision Transformer-Based Unsupervised Anomaly Detection on Industrial Dataset Dec 2024 - Jan 2025

- Constructed a Vision Transformer-based autoencoder using ViT and capsule networks with Gaussian Density Mixture Model, attaining 20% higher AUROC than CNN-based methods on the BeanTech AD dataset

Serverless Matrix Multiplication Using Map-Reduce Aug 2024 - Sept 2024

- Formulated a serverless matrix multiplication framework using Google Cloud with Google Cloud Functions, Cloud Pub/Sub, Cloud Memorystore (Redis and memcached), enabling scalable and parallel processing of matrices from MB to 10 GB scale
- Benchmarked and upgraded performance, achieving a 25% gain in processing efficiency by addressing key bottlenecks

TECHNICAL SKILLS

Certifications: AWS Certified Developer Associate — AWS Certified Cloud Practitioner
Programming Languages: Python, Java, C, C++, JavaScript (Node.js, TypeScript), Tableau
Machine Learning: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, Keras, JAX, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, spaCy, NLTK, LLMs, BERT, GPT, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Pinecone, Weaviate, Chroma)
Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, RDS, CloudFormation), GCP (Cloud Functions, BigQuery, Pub/Sub, Memorystore), Docker, Kubernetes, Apache Spark, Hadoop, CI/CD, Microservices, Serverless
Full-Stack Development: React.js, Next.js, Node.js, Spring Boot, Hibernate, PostgreSQL, MongoDB, Redis, REST, GraphQL, Neo4j